

Swap x_1 with x_4

$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7$$

$$\Rightarrow x_4 = 1.8 - 0.25x_2 + 0.75x_3 - 0.25x_1 + 1.8x_5 + 1.5x_6 + 1.8x_7$$

$$\max_{\mathbf{x}} f(\mathbf{x}) = -1 - 7x_2 - 7x_3 + 5x_4 + 5x_5 + 8x_6 + 3x_7$$

$$\Rightarrow \max_{\mathbf{x}} f(\mathbf{x}) = 7.8 - 8.2x_2 - 3.2x_3 - 1.2x_1 + 14x_5 + 16x_6 + 12x_7$$

$$x_8 = 3 + 6x_2 - 6x_3 + 2x_4 - 7x_5 + 2x_6 + 7x_7$$

$$\Rightarrow x_8 = 6.5 + 5.5x_2 - 4.5x_3 - 0.5x_1 - 3.5x_5 + 5x_6 + 10x_7$$

$$x_9 = 5 - 3x_2 - x_3 - 2x_4 - 7x_5 - 7x_6 - 4x_7$$

$$\Rightarrow x_9 = 1.5 - 2.5x_2 - 2.5x_3 + 0.5x_1 - 10x_5 - 10x_6 - 7.5x_7$$